

AI_DC_TASK_2

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1 Introduction To Transformers

- In the previous task we understood how a Large Language Model converts words into tokens which are then converted into vectors. These vectors have some meaning with similar words being grouped together in a higher dimensional space.
- Transformers are basically a tool that allows us to more accurately convert words to vectors by inferring their meaning through other words.
- For example let us consider the two phrases "Bank of the river" or "Bank of America". Clearly the word Bank has different meanings in the two phrases. Now the job of the transformer is to change the vectors for the word in the two cases so the meaning encoded is different.
- The mechanism in which it works is simple. Each vector is converted into three different vectors:-
 - Query (Q)
 - Key (K)
 - Value (V)
- Here the Query stands for "what am I looking for?", Key represents the information the word contains and Value represents what information the vector should pass on.
- If we analyze the working of this system in the context of the bank example, the word bank is initially initialised with a general meaning. Then it performs a lookup with the other words in the sentence. From this logic the Bank in the first sentence is influenced by the word "river" lending it the context of a geographical feature or landform. In the other sentence it is influence by "America" and takes upon the meaning of a financial institution.
- This manipulation is done by matrix multiplication with certain learned matrices for each task.

- This is the system of "attention" which enables a transformer to more accurately categorize words by using context from within the sentence.